

Lecture 6 Tutorial — Answers

Q1. Definitions

A correlation exists between two variables when the values of one variable are somehow **associated** with the values of the other variable. A linear correlation exists when the paired data can be approximated by a **straight** line.

Q2. Computing from Summary Statistics

Given

$$n = 5, \quad \sum x = 10, \quad \sum y = 20, \quad \sum x^2 = 30, \quad \sum y^2 = 90, \quad \sum xy = 47$$

1.

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}}$$
$$r = \frac{5(47) - 10(20)}{\sqrt{(5(30) - 10^2)(5(90) - 20^2)}} = \frac{35}{\sqrt{50 \cdot 50}} = \frac{35}{50} = 0.70$$

2. The correlation is **positive**.

Q3. Properties of r

1. **T**
2. **F**
3. **F**

Q4. What r Measures

B. r measures the strength of a linear relationship, and it is not designed to measure nonlinear strength.

Q5. Sensitivity to Outliers

C. the value of r

Q6. Critical r and Significance

Given

$$r = 0.947, \quad \text{critical values } \pm 0.666$$

1. **Yes**
2. Since

$$0.947 > 0.666,$$

the correlation is statistically significant.

Q7. Spurious Correlation and Causality

C. does not by itself establish causality

Q8. Regression Coefficients from r

Given

$$r = 0.60, \quad s_x = 2, \quad s_y = 5, \quad \bar{x} = 10, \quad \bar{y} = 100$$

Using

$$b_1 = r \frac{s_y}{s_x}, \quad b_0 = \bar{y} - b_1 \bar{x}$$

1.

$$b_1 = 0.60 \cdot \frac{5}{2} = 1.5$$

2.

$$b_0 = 100 - 1.5(10) = 85$$

3. The regression equation is

$$\hat{y} = 85 + 1.5x$$

Q9. Prediction and Residual

Using

$$\hat{y} = 85 + 1.5x$$

1. When $x = 12$,

$$\hat{y} = 85 + 1.5(12) = 85 + 18 = 103$$

2. If $y = 105$, then

$$\text{Residual} = y - \hat{y} = 105 - 103 = 2$$

3. Since the residual is positive, the point lies **above** the regression line.

Q10. Model Forms

$$\text{A-1,} \quad \text{B-2,} \quad \text{C-3,} \quad \text{D-4,} \quad \text{E-5}$$

That is:

A. Linear $\rightarrow y = a + bx$

B. Logarithmic $\rightarrow y = a + b \ln x$

C. Power $\rightarrow y = ax^b$

D. Quadratic $\rightarrow y = ax^2 + bx + c$

E. Exponential $\rightarrow y = ab^x$

Q11. Logistic Regression

Logistic regression uses

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

1. When $z = 0$,

$$P(Y = 1) = \frac{1}{1 + e^0} = \frac{1}{2}$$

2. When $z = \ln 3$,

$$P(Y = 1) = \frac{1}{1 + e^{-\ln 3}} = \frac{1}{1 + \frac{1}{3}} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

Using the rule “predict class 1 if $P(Y = 1) \geq 0.5$ ”, class 1 is predicted when

$$z \geq 0$$

So the correct answer is:

B. $z \geq 0$